THE NORMAL CLOSURE IS A QUOTIENT OF A TENSOR POWER OF LENGTH THE SEPARABLE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The tensor map is well defined because it is multilinear over K; surjectivity restates that M' is generated as a K-algebra by the sigma_i(L).

Editorial note. The label lemma-normal-closure-tensor-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-separable-degree ([L : K]_s = |Mor_K(L, K-bar)|), lemma-characterize-normal (normality via equal images of embeddings, and its converse), uniqueness of the normal closure (lemma-normal-closure); standard fact: a K-subalgebra of an algebraic closure is a field (domains algebraic over a field are fields)

Source reference. Original reference 09HQ: sources/stacks-fields/pinned-original-source.tex, source lines 2096–2096. Prerequisite pointer invoked at 2403; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HJ: sources/stacks-fields/pinned-original-source.tex, source lines 1968–1968. Prerequisite pointer invoked at 2395; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a finite extension. Let M/L be the normal closure of L over K. Then there is a surjective map*

$$L \otimes_K L \otimes_K \ldots \otimes_K L \longrightarrow M$$

of K-algebras where the number of tensors can be taken $[L : K]_s \leq [L : K]$.

Proof. Choose an algebraic closure $\overline{K}$ of K. Set $n = [L : K]_s = |\operatorname{Mor}_K(L, \overline{K})|$ with equality by Lemma 09HJ. Say $\operatorname{Mor}_K(L, \overline{K}) = \{\sigma_1, \ldots, \sigma_n\}$. Let $M' \subset \overline{K}$ be the K-subalgebra generated by $\sigma_i(L)$, $i = 1, \ldots, n$. Then M' is a field since any K-subalgebra of $\overline{K}$ is a field. Any K-algebra map from M' to $\overline{K}$ permutes the σ_i so sends M' into and onto M'. By construction the field M' is generated by conjugates of elements of $\sigma_1(L)$. Having said this it follows from Lemma 09HQ that M' is normal over K and that it is the smallest normal subextension of $\overline{K}$

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containing $\sigma_1(L)$. By uniqueness of normal closure we have $M \cong M'$. Finally, there is a surjective map

$$L \otimes_K L \otimes_K \ldots \otimes_K L \longrightarrow M', \quad \lambda_1 \otimes \ldots \otimes \lambda_n \longmapsto \sigma_1(\lambda_1) \ldots \sigma_n(\lambda_n)$$

and note that $n \leq [L : K]$ by definition. $\square$